

MICROBIOLOGY

CHAPTER 8

Control of Microorganisms in the Environment



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Reference:

Slides, Dr.Sereen
records, Dr. Abdul
Kareem lectures &
the book

Chapter 8: Control of Microorganisms in the Environment

We're going to learn about: (o~o)

1. Definition of frequently used terms
2. Microbial death
3. Microbial control methods
4. Evaluation of antimicrobial agents effectiveness

🌸 Definition of frequently used terms

By using microbial control methods we will:

Kill the bacteria or Eliminate their number

بهاد التشابتر رح ناخذ عدة طرق للتحكم في البكتيريا وهاي الطرق يا اما بتساعدني بإني اقتلها بشكل نهائي وما يضل اشي منها او بقلل من عدد البكتيريا الموجودة في عينة معينة

Killing → can't find any bacteria alive

Elimination → a few of them remains

في عدة مصطلحات مستخدمة للتفريق بين النتائج الي بتعطيني يها كل طريقة من الطرق وهاي المصطلحات لا لازم نفرق بينهم ونحفظهم منيح وهمة:

- **Sterilization:** destruction or removal of all viable organisms..does not necessarily mean KILLING them. Used on inanimate objects

لما انا اعمل sterilization هاد يعني انو ما ضل عندي بالعينة اي microorganism سواء كان ضار او غير ضار..حتى ال spores بأنواعهم ما بضل منهم اشي، بس هاي العملية لا تعني بالضرورة اني قتلت الكائنات الموجودة المهم انو تمت ازالتهن من العينة المراد تعقيمها. ال sterilization بس يُستخدم على الاشياء الجماد

- **Disinfection:** Definition 1: killing, inhibition, or removal of disease causing (pathogenic) organisms

Definitions 2: the destruction or removal of vegetative pathogens but not bacterial endospores. Usually used only on inanimate objects

Dis- → Removal

➡ ازالة الكائنات المسببة للعدوى

-infection → عدوى

هاد المصطلح يعني اني قضيت على اي مسبب للمرض..سواء بقتله او تثبيطه او ازالته ولكنه ما بقضي على ال endospores عكس ال sterilization الي ما بخلي اشي ابداء

- Disinfectants: Agents (usually chemicals) used for disinfection

ال disinfectants همة المواد الي بستخدمهم عشان اعمل disinfection للجملادات. المنظفات هي مثال على disinfectants زي علبه الكلور منلاقي مكتوب عليها disinfectant.

- **Antisepsis:** prevention of infection of living tissue by microorganisms

هاد المصطلح يُستخدم للدلالة على تعقيم الانسجة الحية فقط

Anti- → مضاد

-sepsis → تقرحات

- Antiseptics : Definition 1: chemical agents that kill or inhibit growth of microorganisms when applied to tissue

Definition 2: chemicals applied to body surfaces to destroy or inhibit vegetative pathogens

→ Because they must not cause too much harm to the host, antiseptics are generally not as toxic as disinfectants

نرجع للمنظفات؛ عندكم مثلا علبة الديتول مكتوب عليها disinfectant and antiseptic يعني بتقدر تستخدمه لتعقيم الاسطح ومختلف الاشياء الجامدة بالإضافة الى انو منقدر تستخدمه لتعقيم الجروح واي نسيج حي. بس علبة الكلور مكتوب عليها disinfectant يعني بس منعقم فيها الجمادات ويمنع استخدامه على الانسجة الحية..حتى دايما بنصحوا انو نلبس قفازات لما يتم التعامل معو..علبة الهايجين مكتوب عليها بس antiseptic..فهلأ صرنا قادرين نفهم ونعرف كل علبة منشترها لشو منقدر نستخدمها (٢٠٠٠)

- **Sanitization**: Reduction of microbial population to levels deemed safe (based on public health standards)
 - Cl^- is added to tap water in a very low amounts that are enough to kill major bacterial cultures and by that reducing their number to a limit that is safe for us (globally)

ال sanitization يقصد فيه تقليل عدد البكتيريا الموجود في عينة معينة للدرجة الي بتخليها آمنة للاستخدام. مية الحنفية بنضافلها كلور..نسبة الكلور هاي ما بتقتل كل البكتيريا الموجودة بالمى ولكن بتقلل من عددها كثير للدرجة الآمنة الناء..عدد البكتيريا الي بتضل متفق عليه عالميا بانها آمنة وما بتسبب امراض للإنسان

- **Chemotherapy**: use of chemicals to kill or inhibit growth of microorganisms within host tissue (internally)

Antimicrobial Agents:

Agents that kill microorganisms or inhibit their growth. They may be:

- -cidal agents: agents that **Kill**
 - **Germicide**: kills pathogens and many nonpathogens but not necessarily endospores
 - include bactericides (بتقتل البكتيريا), fungicides (بتقتل الفطريات), algicides (بتقتل الطحالب), and viricides (بتقتل الفيروسات)
- -static agents: agents that **inhibit growth** (تقليل عدد)
 - include bacteriostatic and fungistatic (انتبهوا او ما في للفيروسات لأنها اساسا مو "كائن" وما بتتكاثر او يزيد عددها لحالها)

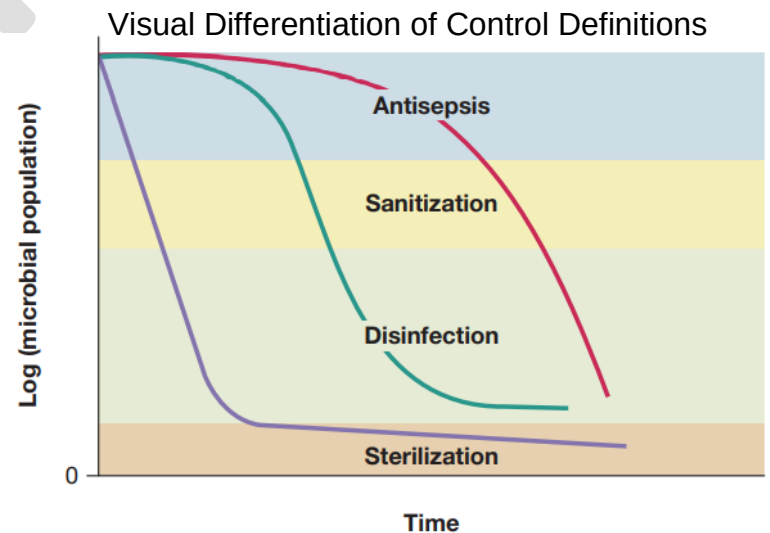


Figure 8.2 Impact of Biocide Exposure. Three exponential plots of survivors versus time of biocide exposure, indicating the potential kinetics of biocide action. Note how the general terms referencing microbial control are reflected by decreasing numbers of microbes. For example, sterilization refers to the absence of viable organisms regardless of biocide kinetics.

عندي مقطعين (suffixes) ببجوا اخر الكلمة مهمين جدا .. لما احكي **bactericide** يعني هاي المادة بتقتل البكتيريا؛ لما احكي **bacteriostatic** يعني هاي المادة بتقلل عدد البكتيريا الموجودة ما بتقتلها. دايمًا بشبهوا هذول المقطعين بالتلاجة والفریزر.. التلاجة ← static والفریزر ← cide من حيث انو الاكل الي بالتلاجة ممكن يقعد مثلاً بس اسبوع وبخرب بعدين بسبب تراكم الكائنات الدقيقة المختلفة عليه ونموها طول هاد الاسبوع ولكن بشكل ابطي طبعاً من انها تكون برا التلاجة. اما لما احط الاكل بالفریزر بضل شهور وممكن سنين حسب نوعه

🌸 Microbial Death

The Pattern of Microbial Death Mirrors the Pattern of Microbial Growth

→ Microorganisms are not killed instantly

→ Population death usually occurs exponentially

يعني زي ما نمو البكتيريا بصير بشكل متدرج مو فجأة ولا مرة وحدة عددها يزيد.. موتها كمان ما بصير فجأة وانما بشكل متدرج على شكل اقتران لوجاريتمي وبصير عددها يقل كل ما الزمن زاد

- Measure of agent's killing efficiency:

- Decimal reduction time (D value): the time required to kill 90% of bacterial populations at a fixed temperature
- Z value : How much would you increase the temperature to reduce d value by 1 log (10 fold)

→ must be sure persister cells (viable but nonculturable condition) are dead

- once they recover they may regain the ability to reproduce and cause infection

ال decimal reduction time: هو الوقت الي محتاجه لنقتل 90% من bacterial population معين عند درجة حرارة معينة (يعني بتثبت درجة الحرارة وبشوف اديه بدي وقت عشان اقل ال 90% من البكتيريا) باستخدام antimicrobial agent معين

ال z value هي الزيادة في درجة الحرارة اللازمة لأنني اقل ال d value تبعت هاد ال agent بمقدار 1 log (او عشر اضعاف)

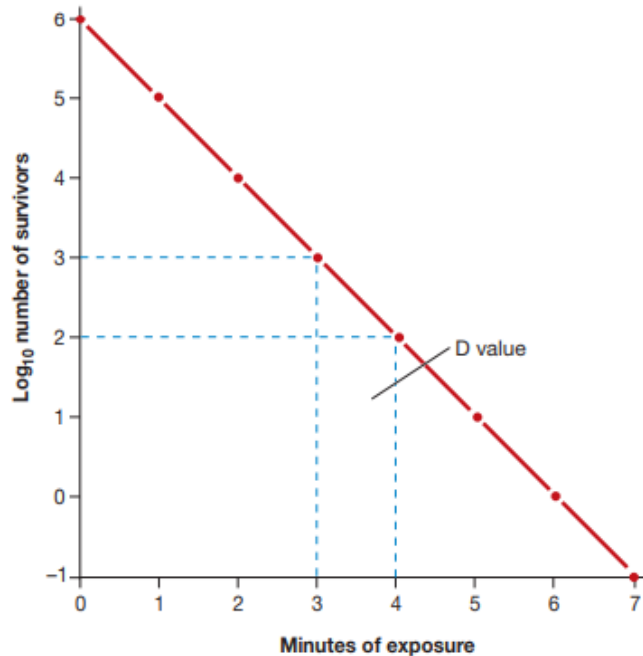
يعني بتقدروا تفكروا فيها بهاي الطريقة.. عشان انا اتخلص من البكتيريا بشكل اسرع بزيد الحرارة (الحرارة عم بتزيد والوقت المستغرق عم يقل) (هذول القيمتين جدا مهمين ركزوا عليهم)

هلاً اذا بتتذكروا بتشابتر 7 حكينا عن انو في كثير مراحل بتمر فيها البكتيريا لتصير ميتة تماماً وانو خلال هاي المراحل بتكون لا عيشة ولا ميتة.. ف عندكم هون عم نحكي عن وجوب قتل 90% من البكتيريا.. طيب كيف بقدرنا يحددوا اذا ماتت او لا ؟ غالباً يتم التعرف على موت الخلية عن طريق ا زراعتها على culture media معين بدغم نموها.. اذا ما نمت عليه وتكاثرت هاد يعني انها ميتة.. هاي هي الطريقة المستخدمة بالعادة ولكن وجدوا مشكلة معينة ولازم يتأكدوا من التخلص منها وهي حالة للبكتيريا منسيميا **(VBNC) Viable But Non-Culturable**؛ بهاي الحالة بتكون البكتيريا عيشة.. بس مؤقتاً مو قادرة تتكاثر وتتضاعف؛ ولما تتوفر لها بعض الظروف المناسبة والمهمة بترجع عادي وبتتكاثر.. زي كأنها كانت نائمة وصحية. في الاختبارات التقليدية لإثبات قدرة antimicrobial agent معين على قتل البكتيريا لازم يتأكدوا من انو كمان قادر يقتل ال VBNC كمان لأنها ممكن ترجع تنشط وتتكاثر وتعمل مشاكل ف المعنى انو الاختبار لأي agent يكون دقيق كثير ولازم حريفاً يتم التخلص من البكتيريا ونكون متأكدين انها ما رح ترجع ابدًا.

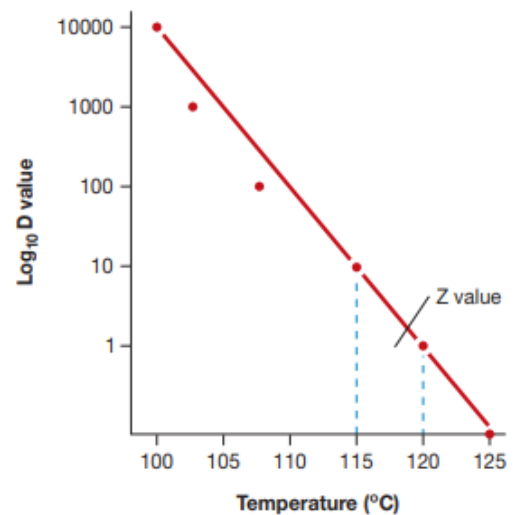
Table 8.1 A Theoretical Microbial Heat-Killing Experiment

Minute	Microbial Number at Start of Minute	Microorganisms Killed in 1 Minute (90% of Total) ¹	Microorganisms at End of 1 Minute	Log ₁₀ of Survivors
1	10 ⁶	9 × 10 ⁵	10 ⁵	5
2	10 ⁵	9 × 10 ⁴	10 ⁴	4
3	10 ⁴	9 × 10 ³	10 ³	3
4	10 ³	9 × 10 ²	10 ²	2
5	10 ²	9 × 10 ¹	10	1
6	10 ¹	9	1	0
7	1	0.9	0.1	-1

¹ Assume that the initial sample contains 10⁶ vegetative microorganisms per milliliter and that 90% of the organisms are killed during each minute of exposure. The temperature is 121°C.



(a)



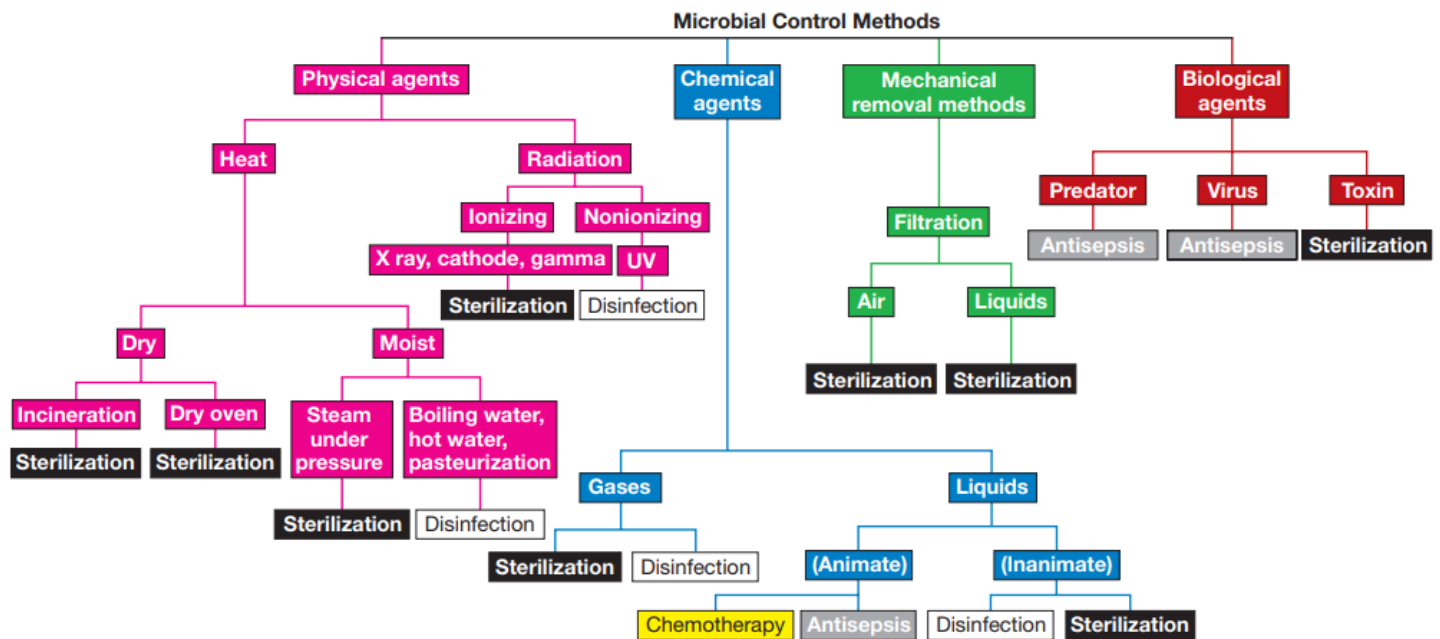
(b)

Figure 8.3 The Pattern of Microbial Death. (a) An exponential plot of the survivors versus the minutes of exposure to heating at 121°C. In this example, the D value is 1 minute. The data are from table 8.1. (b) An exponential plot of D values versus temperature. The temperature change at a given D value that reduces the population by one log unit is known as the Z value.

🌸 Microbial control methods

Antimicrobial agents can be chemical, physical, mechanical, and biological

- Mechanical agents → mainly by the filtration of air or liquids
- Physical agents → uses Heat or Radiation
- Chemical agents → using gases or liquid substances
- Biological agents → uses : Predators, Viruses or Toxins



Disinfection: The destruction or removal of vegetative pathogens but not bacterial endospores. Usually used only on inanimate objects.

Sterilization: The complete removal or destruction of all viable microorganisms. Used on inanimate objects.

Antisepsis: Chemicals applied to body surfaces to destroy or inhibit vegetative pathogens.

Chemotherapy: Chemicals used internally to kill or inhibit growth of microorganisms within host tissues.

هـاـد الشـكـل بـلـخـص كـل التـشـابـتـر ضـلـكـم مـتـبـعـيـن مـعـو بـكـل agent لانـو جـدا جـدا مـهـم (▽) (▽) (▽)

A. Mechanical agents

The main idea is that they rely on "barriers" → Filters

- They reduces microbial population or do sterilization by removing these populations ..not killing them
- Usually used for heat-sensitive solutions (like protein and hormone solutions)
- used to reduce microbial populations in air and liquids

العوامل الميكانيكية بتعتمد بشكل مباشر على مبدأ الفلترية. يكون عندي سائل او غاز معين مراد تعقيمه ف بمرقه من خلال هاد الفلتر الي رح يحبس ال microorganisms عليه ويطلع السائل او الغاز خالي من اي microorganism..لاحظوا انو تخلصنا منهم عن طريق ازالتهم مو قتلهم

- Filtering liquids: we use Membrane filters (figure 8.5): porous membranes with defined pore sizes (0.2 μm in diameter) and a thickness of 0.1 mm; made of cellulose acetate, cellulose nitrate, polycarbonate, polyvinylidene fluoride, or other synthetic materials. They remove microorganisms primarily by physical screening → microorganisms will be trapped at the surface of the membrane → the only thing that can pass through these filters is the virus (remember..it is measured in Nano-meters)

- Filtering air

Some examples:

- Surgical masks
- Cotton plugs on culture vessels
- High-efficiency particulate air (HEPA) filters: used in laminar flow: biological safety cabinets (see figure 8.6)

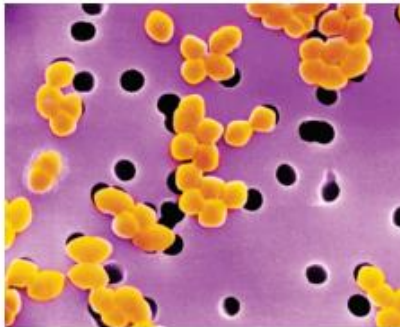
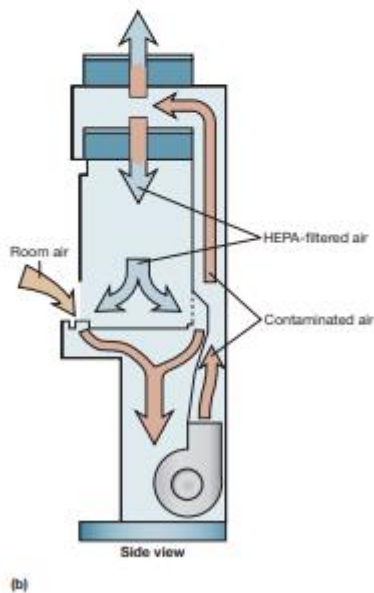


Figure 8.5 Membrane Filter. *Enterococcus faecalis* resting polycarbonate membrane filter with 0.4 µm pores (×5,900).



(a)



(b)

Figure 8.6 A Biological Safety Cabinet. (a) A technician pipetting potentially hazardous material in a safety cabinet. (b) A schematic diagram showing the airflow pattern within a class II safety cabinet.

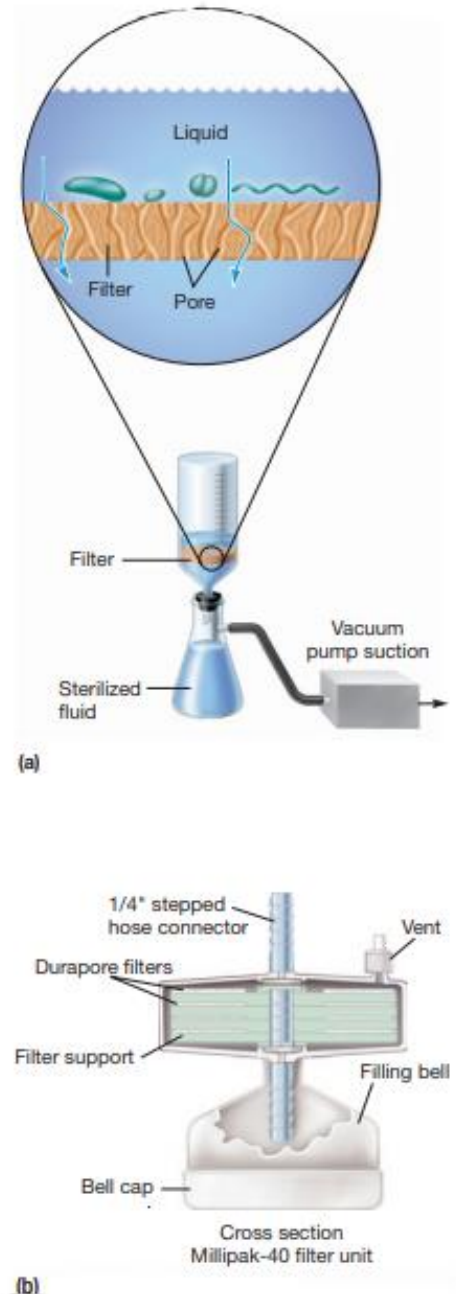
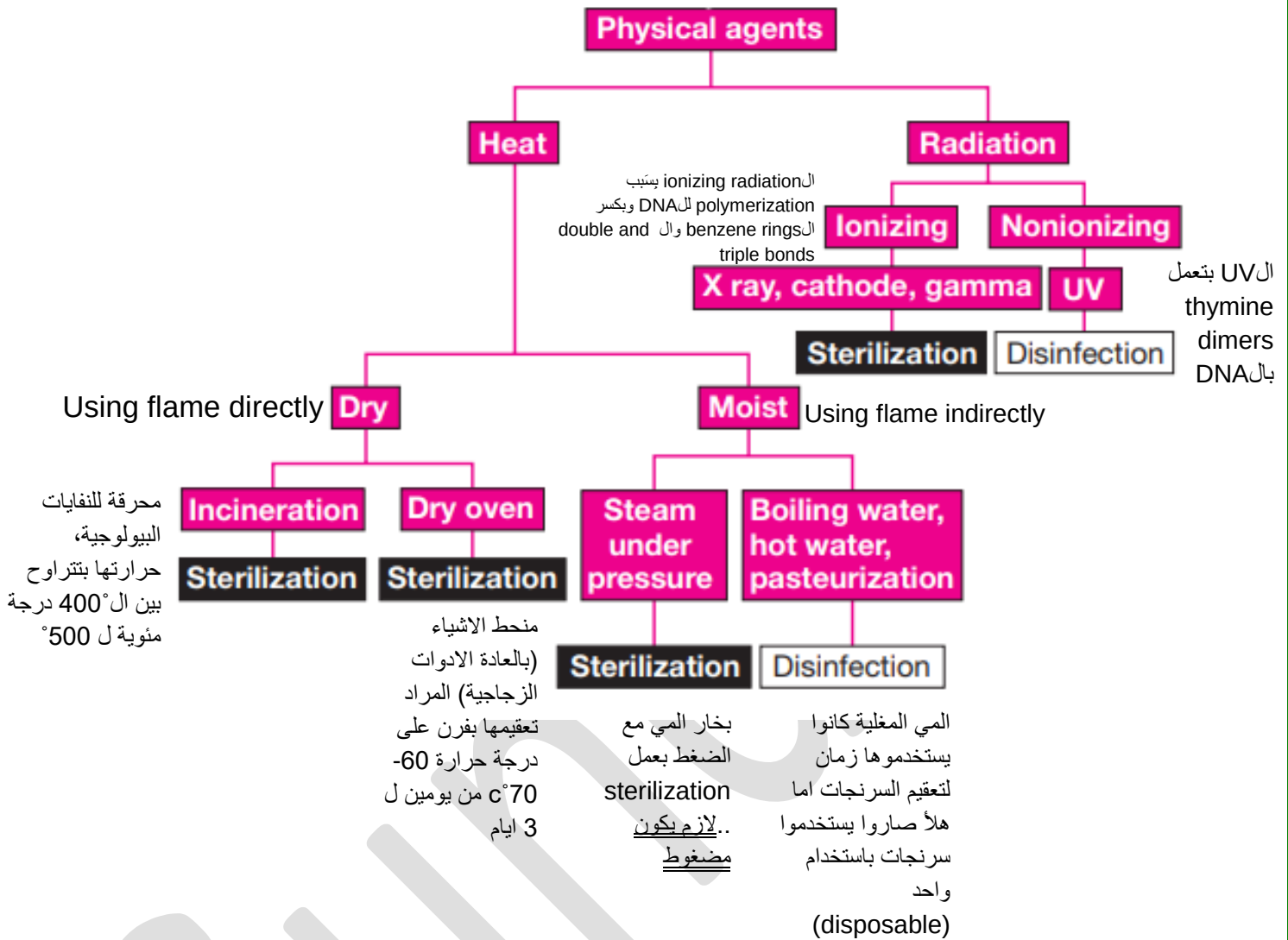


Figure 8.4 Membrane Filter Sterilization. The liquid to be sterilized is pumped through a membrane filter and into a sterile container. (a) Schematic representation of a membrane filtration setup that uses a vacuum pump to force liquid through the filter. The inset shows a cross section of the filter and its pores, which are too small for microbes to pass through. (b) Cross section of a membrane filtration unit. Several membranes are used to increase its capacity.

B. Physical agents:



١/ (ω) هلا رح نبش نشرح كل وحدة فيهم بالتفصيل

I. Heat

a- Moist Heat:

- Destroys viruses, fungi, and bacteria
- Boiling does not destroy spores and does not sterilize (remember in sterilizing, even spores should be killed)
- It degrades nucleic acids, denatures proteins, and disrupts membranes

Table 8.2 Approximate Conditions for Moist Heat Killing		
Organism	Vegetative Cells	Spores
Yeasts	5 minutes at 50–60°C	5 minutes at 70–80°C
Molds	30 minutes at 62°C	30 minutes at 80°C
Bacteria ¹	10 minutes at 60–70°C	2 to over 800 minutes at 100°C 0.5–12 minutes at 121°C
Viruses	30 minutes at 60°C	

¹ Conditions for mesophilic bacteria. انتبهوا انو درجة الحرارة بتزيد لقتل الـ spores

ولاحظوا عند الـ spores تبكون البكتيريا قل الوقت لما زدت درجة الحرارة وعرضتها لضغط والفرق جدا كبير.. من 800 دقيقة لـ 0.5-12 دقيقة!

- Steam Sterilization

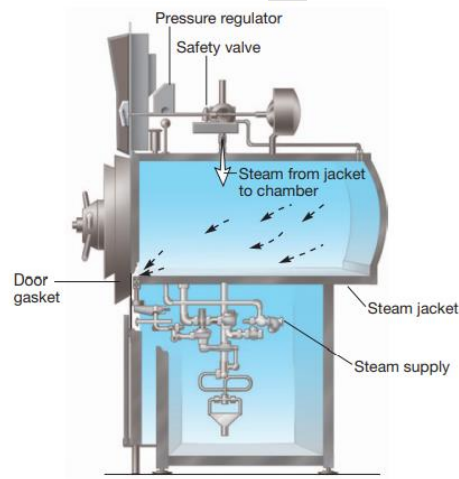
Carried out above 100°C which requires saturated steam under pressure

- Uses an autoclave (121°C and 15 pounds per sq. in. (psi) pressure)
- Effective against **all types of microorganisms (including spores!)**
- Quality control - includes strips with *Geobacillus stearothermophilus*

عشان يتأكدوا من عملية الsterilization وانو فعلا ما ضل ولا اي نوع بكتيريا بحطوا مع المواد المراد تعقيمها spores لبكتيريا اسمها *Geobacillus stearothermophilus* وهاي البكتيريا عندها مقاومة جدا عالية للحرارة (thermophilic) فيستخدموها كمقياس.. اذا حطوها بالautoclave مع المواد الي بدهم يعقموها وبعدين زرعوها على plate وما عاشت وما اعطت cultures هاد يعني انو عملية الsterilization نجحت وفعلا ما ضل ولا اشي



(a)



(b)

Figure 8.7 The Autoclave. (a) A modern, automatically controlled autoclave or sterilizer. (b) Longitudinal cross section of a typical autoclave showing some of its parts and the pathway of steam.

- Pasteurization → used for liquids

- Controlled heating at temperatures well below boiling
- Used for milk, beer, and other beverages
- Process **does not sterilize** but does kill pathogens present and slow spoilage by reducing the total load of organisms present

البسترة فكرتها الأساسية اني اقضي على كل الpathogens او مسببات الامراض الموجودة في سائل معين بالإضافة الى انها بتقلل العدد الكلي للبكتيريا الموجودة بس ما بتروح بشكل نهائي والدليل انو مثلا الحليب بعد فترة من الزمن بخرب نتيجة لتضاعف البكتيريا وتكاثرها بس بنفس الوقت قليل ما نسمع انو في حد صار معو السيل البقري (tuberculosis) بعد ما شرب حليب مبستر لانو زي ما حكينا البسترة بتقضي على الpathogens ولكن في انواع تانية ما بتسبب امراض بتضل

بالبسترة يتم استخدام درجات حرارة اقل من الغليان بشوي.. اول شي كانت على درجة حرارة 60° لمدة 30 دقيقة ولكنهم وجدوا انو البكتيريا الي بتسبب السيل ضلّت عايشة فرفعوها ل63° وبرضو رجعوا وجدوا انو في بكتيريا اسمها *Coxiella* الي بتسبب الQ fever ضلّت عايشة فرفعوا الحرارة ل65° لمدة 30 دقيقة.

اخر طريقة وجدوها للبسترة اسمها The Flash method أو High-Temperature Short-time (HTST) يتم على درجة حرارة 72°C لمدة 15 ثانية

بالنسبة للـ **heat** بشكل عام ف في **two factors** مهمين بكل طريقة من الطرق الي حكيها او رح نحكي عنها .. همة درجة الحرارة والوقت .. هدول همة الي يتم التلاعب فيهم .. يعني مثلاً زي ما شفا بالصفحة الي قبل وبالـ **z value** اذا بزيد درجة الحرارة بقل الوقت الي بحتاجه. فمثلاً اذا كان عندك سائل معين حساس للحرارة زي البروتينات و هيك .. بتروح بتقلل درجة الحرارة وبتزيد الوقت عشان ما يصير **denaturing** للبروتينات

بعض المواد ما بتقدر تصمد بالـ **autoclave** ، ووجود الـ **spores** بمنع استخدام اي طريقة ثانية لتعقيمها. بالنسبة لهاي المواد في عملية يتم فيها تعقيم متقطع او **intermittent sterilization** ، اسمها **Tyndallization**. ركزوا معي هلاً * ☆ — (n^{II} - II)

بهاي العملية بعرض العينة لبخار درجة حرارته عالية لمدة 30 دقيقة بعدين بعمل **incubation** لـ 24 ساعة وبعيد هاي العملية 3 مرات. الفكرة منها انو لما اعرض البكتيريا للبخار الساخن لمدة 30 دقيقة .. البكتيريا العائشة او الـ **viable** رح تموت وبنفس الوقت اذا في **spores** رح يتم تنشيطهم (مو قتلهم) (تذكروا الـ **spores** بتحولوا لـ **viable cells** عن طريق الحرارة)؛ هلاً هدول الـ **spores** الي تم تنشيطهم مع الـ **incubation** رح يصيروا **viable cells**. ثاني يوم يرجع بعرضهم لكان بخار ساخن ويقتل الـ **viable cells** الـ اساسا اصلهم **spores** وازا كان في ضايل **spores** برضو رح يتم تنشيطهم. بتتعد العملية 3 مرات عشان نتأكد انو ما ضل **spores** ما صارلهم **activation** وماتوا و هيك نكون متأكدين انو تم تعقيم العينة (تذكروا انو البخار الساخن لحالو بدون الضغط ما بعمل **sterilization** وما يقتل الـ **spores** لهيك بنشطهم بالأول بعدين بقتلهم (o^o - o^o))

b- Dry heat

- Less effective than moist heat sterilization, requiring higher temperatures and longer exposure times
→ items subjected to 160-170°C for 2 to 3 hours
- How does it kill? It Oxidizes cell constituents and denatures proteins

❖ The spores of *Clostridium botulinum*, the cause of botulism (تستخدم كسلاح بيولوجي), are killed in 5 minutes at 121°C by moist heat but only after 2 hours at 160°C by dry heat. However, dry heat has some definite advantages.

- Dry Heat Incineration

- Bench top incinerators are used to sterilize inoculating loops used in microbiology laboratories

• Note that dry heat sterilization and dry heat incineration are different control methods!

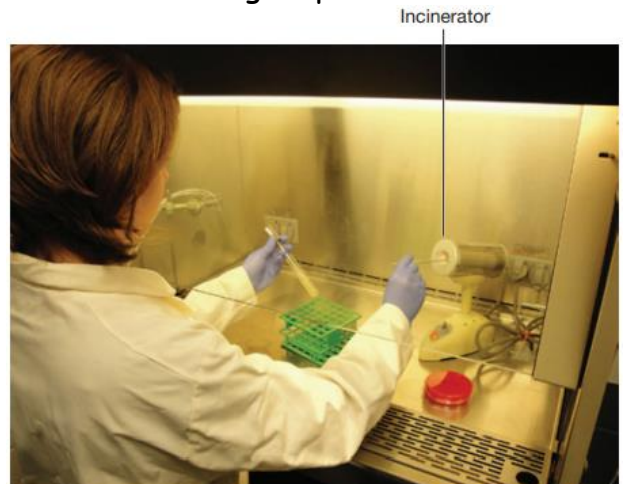


Figure 8.8 Dry Heat Incineration. Bench-top incinerators are routinely used to sterilize inoculating loops used in microbiology laboratories.

II. Radiation

a- Nonionizing radiation→Ultraviolet (UV) Radiation

- Wavelength of **260** is most bactericidal because it is absorbed by the DNA
- Causes **thymine dimers** preventing replication and transcription
- UV limited to surface sterilization because it does not penetrate glass, dirt films, water, and other substances and because of that it is used as a sterilizing agent only in a few specific situations
- Has been used for water treatment

اكتر شي يتم استخدام ال UV فيه هو غرف العمليات وال laminar flow cabinet عشان يعقموهم..طبعا لما بييجوا يعقموا غرف العمليات ما يكون في اي حد جوا لأنو بحرق الجلد وبضر بالعيون

b- Ionizing radiation→Gamma, Cathode & X-ray

- Gamma radiation penetrates deep into objects
- it is very effective→can destroys bacterial endospores; but it's not always effective against viruses
- Used for sterilization and pasteurization of antibiotics, hormones, sutures, plastic disposable supplies, and food(it uses cobalt 60 that affects germs but not the food)

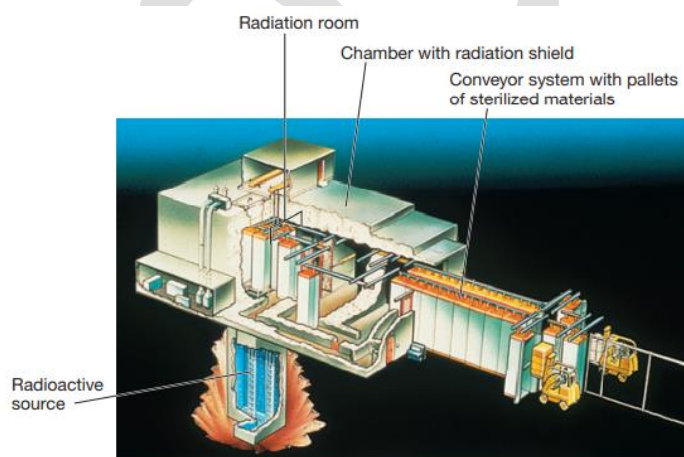


Figure 8.9 Sterilization with Ionizing Radiation. An irradiation machine that uses radioactive cobalt 60 as a gamma radiation source to sterilize fruits, vegetables, meats, fish, and spices.

C. Chemical agents

هالأ بدنا نعرف كل مادة شو لازم يكون متوفر فيها عشان مثلا اقدر احكي انها بتعمل disinfection او مثلا antiseptis بعدين رح نحكي عن ال chemical agents سو يلا.. (((• ω •) \ (

• Chemical Control Agents and Their Requirements

Depends on three factors:

- Type of the microorganism
- Concentration of the chemical material
- Length of exposure

1. Disinfection

must be effective against wide variety of infectious agents at low concentrations

2. Antisepsis

Antiseptics should reduce the number of pathogens on human tissue to prevent infection.

One potentially serious problem is the overuse of antiseptics; it caused resistance to the antibacterial agent triclosan and as a result selected for triclosan resistant bacteria and may possibly select for antibiotic resistant bacteria

3. Sterilization

→ Whichever chemical is used, it must be effective in the presence of organic matter and should be stable in storage

- Activity Levels of Different Chemicals

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Table 8.3 Activity Levels of Selected Biocides		
Class	Use Concentration of Active Ingredient	Activity Level ¹
Gas		
Ethylene oxide	450–500 mg/L ²	High
Liquid		
Glutaraldehyde, aqueous	2–8%	High to intermediate
Formaldehyde + alcohol	8 + 70%	High to intermediate
Stabilized hydrogen peroxide	6–30%	High to intermediate
Formaldehyde, aqueous	6–8%	High to intermediate
Iodophors, high concentration	5,000–10,000 mg/L ³	High to intermediate
Iodophors, low concentration	75–150 mg/L ³	Intermediate to low
Iodine + alcohol	0.5 + 70%	Intermediate
Chlorine compounds	500–5,000 mg/L ⁴	Intermediate
Phenolic compounds, aqueous	0.5–3%	Intermediate to low
Iodine, aqueous	1%	Intermediate
Alcohols (ethyl, isopropyl)	62–70%	Intermediate
Quaternary ammonium compounds	0.1–0.2% aqueous	Low
Chlorhexidine	0.75–4%	Low
Hexachlorophene	1–3%	Low
Mercurial compounds	0.1–0.2%	Low

¹ High-level disinfectants destroy vegetative bacterial cells including *M. tuberculosis*, bacterial endospores, fungi, and viruses. Intermediate-level disinfectants destroy all of the above except spores. Low-level agents kill bacterial vegetative cells except for *M. tuberculosis*, fungi, and medium-sized lipid-containing viruses (but not bacterial endospores or small, nonlipid viruses).

² In autoclave-type equipment at 55 to 60°C.

³ Available iodine.

⁴ Free chlorine.

Source: From Seymour S. Block, DISINFECTION, STERILIZATION AND PRESERVATION.

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- Efficiency of Different Chemical Compounds

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Class	Disinfectant ¹	Antiseptic	Comment
Gas			
Ethylene oxide	3–4	0	Sporicidal; toxic; good penetration; requires relative humidity of 30% or more; microbicidal activity varies with apparatus used; absorbed by porous material; dry spores highly resistant; presoaking is most desirable
Liquid			
Glutaraldehyde, aqueous	3	0	Sporicidal; active solution unstable; toxic
Stabilized hydrogen peroxide	3	0	Sporicidal; solution stable up to 6 weeks; toxic orally and to eyes; mildly skin toxic; little inactivation by organic matter
Formaldehyde + alcohol	3	0	Sporicidal; noxious fumes; toxic; volatile
Formaldehyde, aqueous	1–2	0	Sporicidal; noxious fumes; toxic
Phenolic compounds	3	0	Stable; corrosive; little inactivation by organic matter; irritates skin
Chlorine compounds	1–2	0	Fast action; inactivation by organic matter; corrosive; irritates skin
Alcohol	1	3	Rapidly microbicidal except for bacterial spores and some viruses; volatile; flammable; dries and irritates skin
Iodine + alcohol	0	4	Corrosive; very rapidly microbicidal; causes staining; irritates skin; flammable
Iodophors	1–2	3	Somewhat unstable; relatively bland; staining temporary; corrosive
Iodine, aqueous	0	2	Rapidly microbicidal; corrosive; stains fabrics; stains and irritates skin
Quaternary ammonium compounds	1	0	Inactivated by soap and anionics; compounds absorbed by fabrics; old or dilute solution can support growth of Gram-negative bacteria
Hexachlorophene	0	2	Insoluble in water, soluble in alcohol; not inactivated by soap; weakly bactericidal
Chlorhexidine	0	3	Soluble in water and alcohol; weakly bactericidal
Mercurial compounds	0	±	Greatly inactivated by organic matter; weakly bactericidal

¹ Subjective ratings of practical usefulness in a hospital environment: 4 is maximal usefulness; 0 is little or no usefulness; ± signifies that the substance is sometimes useful.

Source: From Seymour S. Block, *DISINFECTION, STERILIZATION AND PRESERVATION*.

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- Chemical agents can either be liquids or gases

I. Liquids

- Examples on chemical agents:

1- Phenolics

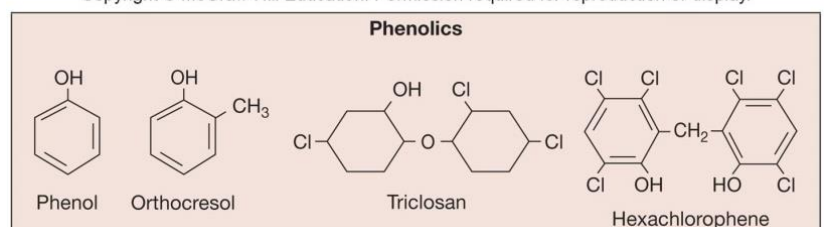
- Commonly used as laboratory and hospital disinfectants and it is the first disinfectant used in hospitals
- Act by denaturing proteins and disrupting cell membranes
- Phenolics are **Tuberculocidal**: Kills *Mycobacterium tuberculosis*, causal agent of tuberculosis. They are effective in presence of organic material, and remain active on surfaces long after application

- Disadvantages:

Disagreeable odor and can cause skin irritation

→ because of that they are not used anymore

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2- Alcohols

• Among the most widely used disinfectants and antiseptics

• Two most common are:

ethanol and isopropanol

• Bactericidal, fungicidal, BUT NOT SPORICIDAL

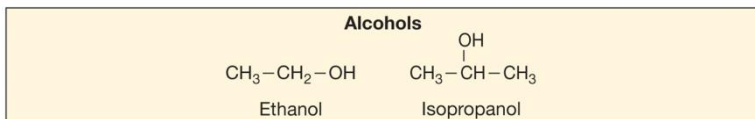
• They inactivate some viruses

• they act by denaturing proteins and possibly by dissolving membrane lipids

• the concentration of alcohols must be between 60 and 80%

→ if higher, it won't be effective

الكحول لازم يكون تركيزه من 60-80% عشان يكون عندي من 20-40% مي بالمحلول وهاي المي مهمة عشان يقدر يفوت الكحول لجوا الخلية عن طريق الosmosis ويقتل البكتيريا

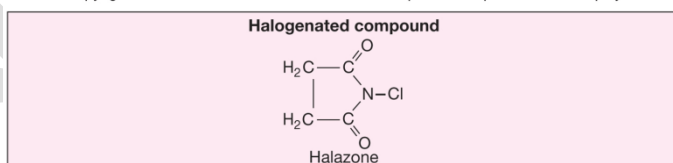


3- Halogens

• Any of five elements: Fluorine, Chlorine, Bromine, Iodine, and Astatine

• Important antimicrobial agents

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- Iodine:

- Used as a Skin antiseptic
- Oxidizes cell constituents and iodinate proteins
- At high concentrations may kill spores (☆w☆)
- Disadvantages: Skin damage, Staining left, and allergies can be a problem
- Iodophore - iodine complexed with organic carrier
 - water soluble, stable, and nonstaining
 - they release iodine slowly to minimize skin burns and irritation

- Chlorine:

- It oxidizes cell constituents
- Important in disinfection of water supplies and swimming pools, used in dairy and food industries, effective household disinfectant
- Destroys vegetative bacteria and fungi
- Chlorine gas is sporicidal
- Can react with organic matter to form carcinogenic compounds

4- Heavy Metals

- e.g. ions of Mercury, Silver, Arsenic, Zinc, and Copper
- Effective but usually toxic

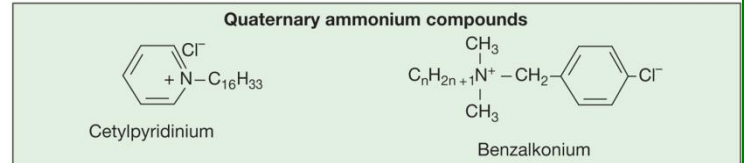
- Combine with and inactivate proteins; may also precipitate proteins

5- Quaternary Ammonium Compounds

- They are detergents(منظفات) that have antimicrobial activity and are effective disinfectants used for decontamination purposes

- Detergents are organic cleansing agents that are amphipathic, having both polar hydrophilic and nonpolar hydrophobic components.
- Cationic detergents are effective disinfectants
- They kill most bacteria, but NOT *M. tuberculosis* or endospores
- They are safe and easy to use, but they are inactivated by hard water and soap

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6- Aldehydes

Commonly used agents are formaldehyde and glutaraldehyde(commonly used in hospitals)

- Highly reactive molecules
- **Sporicidal** and can be used as chemical sterilants
- Combine with and inactivate nucleic acids and proteins

- Effects of Glutaraldehyde figure (8.11):

In Gram Negative:

It polymerizes and then combines with amino acids causing inactivation for protein

In Gram Positive:

It polymerizes and then combines with peptidoglycan which prevents them from participating in other chemical reactions such as those involved in peptidoglycan synthesis

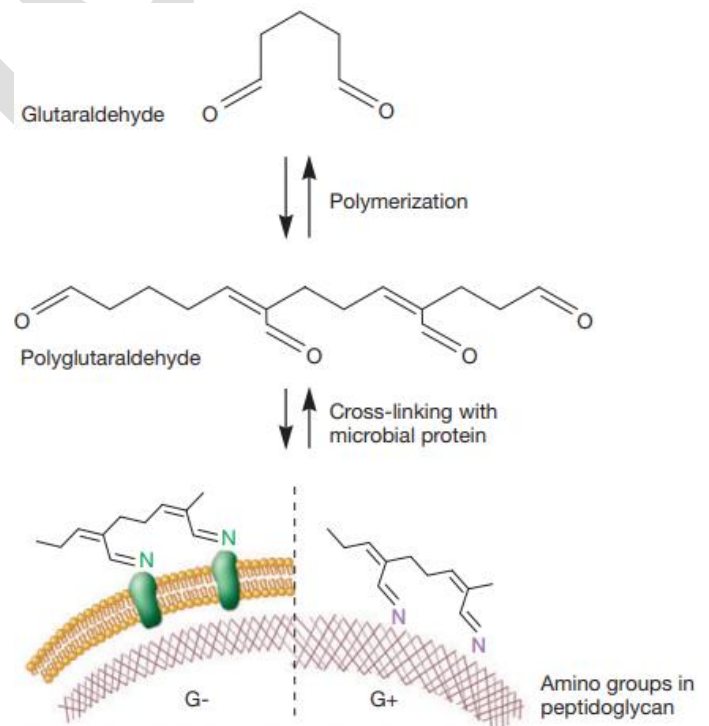


Figure 8.11 Effects of Glutaraldehyde. Glutaraldehyde polymerizes and then interacts with amino acids in proteins (left) or in peptidoglycan (right). As a result, the proteins are alkylated and cross-linked to other proteins, which inactivates them. The amino groups in peptidoglycan are also alkylated and cross-linked, which prevents them from participating in other chemical reactions such as those involved in peptidoglycan synthesis.

II. Sterilizing Gases

→ Used to sterilize heat-sensitive materials

• Microbicidal and sporicidal

- a- Ethylene oxide sterilization is carried out in equipment resembling an autoclave and it is used widely on plastics like Petri dishes and syringes, heart-lung machine components, sutures, and catheters
 - kills by reacting with functional groups of DNA and proteins to block replication and enzymatic activity. It is a particularly effective sterilizing agent because it rapidly penetrates packing materials, even plastic wraps.
 - Sterilization is carried out in an ethylene oxide sterilizer, which resembles an autoclave in appearance. It controls the EtO concentration, temperature, and humidity (figure 8.12).
- b- Betapropiolactone (BPL) :
More effective than ethylene oxide because it is decomposed and degraded faster but ethylene oxide is used more since BPL is carcinogenic
- c- Vaporized hydrogen peroxide

منحط كل الأشياء المراد تعقيمها جوا هاد ال sterilizer ومنشغلو.. ببلش ال ethylene oxide يطلع ويعقم ولازم ننتبه انو لما يكمل تعقيم ..لازميتم ضخ الغاز برا ال chamber وادخال هوا عادي عليه بعدين منتفحو لانو مضر كثير

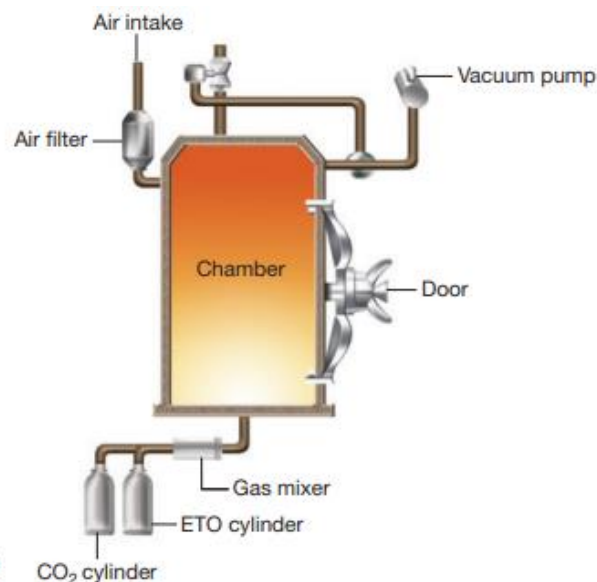


(a)

Figure 8.12 An Ethylene Oxide Sterilizer.

(a) An automatic ethylene oxide (EtO) sterilizer.

(b) Schematic of an EtO sterilizer. Items to be sterilized are placed in the chamber, and EtO and carbon dioxide are introduced. After the sterilization procedure is completed, the EtO and carbon dioxide are pumped out of the chamber and air enters.



(b)

D. Biological agents

- Biological Control of Microorganisms is an emerging field showing great promise
- They are natural control mechanisms that use:
 1. predation by *Bdellovibrio*

الـ *Bdellovibrio* هي بكتيريا بتفترس على الـ *E.coli* ف ممكن يستخدموها للقضاء على الـ *E.coli* بأماكن تواجدها

2. Viral-mediated lysis using pathogen specific bacteriophage lysins (or bacteriophages themselves, such as in a spray format recently approved by the FDA for use on food products)

ممكن يتم استخدام الفيروسات للقضاء على البكتيريا لأنها بتقدر تعمل lysis للبكتيريا

3. toxin-mediated killing using bacteriocins

ممكن نستخدم الـ toxins نفسها تبعون خلية بكتيرية معينة للقضاء على وحدة ثانية

🌸 Evaluation of antimicrobial agents effectiveness

➔ It is a complicated process regulated by U.S. federal agencies

- Environmental Protection Agency
- Food and Drug Administration

- Conditions Influencing the Effectiveness of Antimicrobial Agent Activity:

- Population size
 - larger populations take longer to kill than smaller populations
- Population composition
 - microorganisms differ markedly in their sensitivity to antimicrobial agents and it differs whether we're dealing with vegetative cells or spores
- Concentration or intensity of an antimicrobial agent
 - usually higher concentrations kill more rapidly
 - relationship is not linear (like in alcohols)
- Duration of exposure
 - longer exposure → more organisms killed
- Temperature
 - higher temperatures usually increase killing
- Local environment
 - pH, viscosity, concentration of organic matter, etc. can profoundly impact effectiveness
 - organisms in biofilms are less susceptible to many antimicrobial agents

- Three tests are used:

1. Phenol Coefficient Test

- potency of a disinfectant is compared to that of phenol

منخفض الphenol والchemical المراد استخدامه ومنحطهم عالبكتيريا ومنقارن فعالية هاد الchemical بالphenol

- useful for screening but may be misleading

2. Use dilution test

- determines rate at which selected bacteria are destroyed by various chemical agents

بجربوا تراكيز مختلفة من هاد الchemical وبشوفوا كل واحد منهم كيف بآثر على البكتيريا

3. Normal in-use testing

- testing done using conditions that approximate normal use of disinfectant

The End

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